

Experiment No 2

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Class-D15A Roll no-13

AIM-Study of various types of sensors

1. Temperature Sensor

- **Purpose:** Measures temperature, often in physical environments or processes.
- **Examples in Real World:**
 - DS18B20
- **Key Principle of Working:** Temperature sensors measure temperature using physical properties that change with temperature (like resistance or voltage).
 - Thermistor (resistance changes with temperature)
 - Thermocouple (voltage changes with temperature)
 - RTD (Resistance Temperature Detector) (resistance changes)
- **Key Technologies:**
 - Thermistors, Thermocouples, RTDs
- **Applications:**
 - Climate control (HVAC systems)
 - Industrial process monitoring
 - Home automation (e.g., smart home thermostats)
 - Environmental monitoring

2. Humidity Sensor

- **Purpose:** Measures the humidity (moisture level) in the air.
- **Examples in Real World:**
 - DHT11
- **Key Principle of Working:** The sensor detects the amount of water vapor in the air, often using electrical resistance or capacitance that changes with humidity.
 - Capacitive (change in capacitance with humidity)
 - Resistive (change in resistance with humidity)

- **Key Technologies:**
 - Capacitive, Resistive, and Thermal conductivity
- **Applications:**
 - Weather forecasting
 - Climate control in buildings
 - Agricultural monitoring (greenhouses)
 - Industrial air quality control

3. Gas Sensor

- **Purpose:** Detects the presence and concentration of gases.
- **Examples in Real World:**
 - MQ-2
- **Key Principle of Working:** Gas sensors detect changes in electrical conductivity or other properties when exposed to specific gases.
 - Electrochemical sensors (detect specific gases through chemical reactions)
 - Semiconductor sensors (changes in resistance)
 - Infrared sensors (absorption of IR light by gases)
- **Key Technologies:**
 - Electrochemical, Infrared, Semiconductor
- **Applications:**
 - Environmental monitoring (air quality)
 - Industrial safety (gas leak detection)
 - Automotive (exhaust gas analysis)
 - Home safety (smoke, CO detectors)

4. Motion Sensor

- **Purpose:** Detects movement in a given area.
- **Examples in Real World:**
 - PIR (Passive Infrared) Sensor
- **Key Principle of Working:** Motion sensors detect physical movement, often by detecting infrared radiation (heat) or changes in electromagnetic fields.
 - Passive Infrared (PIR) (detects infrared radiation from objects)

- Ultrasonic sensors (measure distance changes from sound waves)
- **Key Technologies:**
 - Infrared, Ultrasonic, Microwave
- **Applications:**
 - Security and surveillance
 - Smart home lighting
 - Occupancy detection
 - Robotics and automation

5. Light Sensor

- **Purpose:** Measures the intensity of light.
- **Examples in Real World:**
 - BH1750
- **Key Principle of Working:** Light sensors work by detecting the amount of light and converting it to an electrical signal.
 - Photodiodes (converts light to current)
 - LDR (Light Dependent Resistor) (changes resistance with light intensity)
- **Key Technologies:**
 - Photodiodes, LDR, Phototransistors
- **Applications:**
 - Streetlight control
 - Photography/camera systems
 - Solar energy systems
 - Environmental lighting systems

6. Proximity Sensor

- **Purpose:** Detects the presence or absence of an object or its distance from the sensor.
- **Examples in Real World:**
 - HC-SR04 (Ultrasonic Sensor)
- **Key Principle of Working:** Proximity sensors detect objects by emitting a signal (light, sound, or electromagnetic waves) and measuring the return or change in signal.

- Capacitive (detects change in capacitance)
- Inductive (detects metal objects)
- Ultrasonic (measures distance based on sound waves)
- **Key Technologies:**
 - Capacitive, Inductive, Ultrasonic
- **Applications:**
 - Industrial automation
 - Automotive parking sensors
 - Consumer electronics
 - Touchless control systems

7. Pressure Sensor

- **Purpose:** Measures the pressure of gases or liquids.
- **Examples in Real World:**
 - BMP180
- **Key Principle of Working:** Pressure sensors measure the force exerted by a gas or liquid on a surface and convert it into an electrical signal.
 - Piezoelectric (changes voltage under pressure)
 - Capacitive (change in capacitance with pressure)
 - Strain gauge (measures strain due to pressure)
- **Key Technologies:**
 - Piezoelectric, Strain gauges, Capacitive
- **Applications:**
 - Automotive (tire pressure monitoring)
 - Aerospace (altitude measurement)
 - Medical (blood pressure monitoring)
 - Industrial applications (hydraulic and pneumatic systems)

8. pH Sensor

- **Purpose:** Measures the acidity or alkalinity of a solution.
- **Examples in Real World:**
 - PH-4502C

- **Key Principle of Working:** pH sensors work by measuring the concentration of hydrogen ions in a solution, which correlates to pH.
 - Glass electrodes (measure pH via voltage change)
 - Ion-selective field-effect transistors (ISFET) (electrical response to ion concentration)
- **Key Technologies:**
 - Glass electrode, ISFET
- **Applications:**
 - Water treatment and environmental monitoring
 - Food and beverage industry
 - Agriculture (soil pH testing)
 - Laboratories (research and analysis)

9. Flow Sensor

- **Purpose:** Measures the flow rate of liquids or gases.
- **Examples in Real World:**
 - YF-S201 (Water Flow Sensor)
- **Key Principle of Working:** Flow sensors detect the speed or quantity of flow through a pipe or a duct.
 - Turbine flow meters (rotating turbine detects flow)
 - Electromagnetic flow meters (detects flow by induced voltage)
 - Ultrasonic flow meters (uses sound waves to measure flow speed)
- **Key Technologies:**
 - Turbine, Electromagnetic, Ultrasonic
- **Applications:**
 - Industrial process control
 - Oil and gas pipelines
 - Water supply monitoring
 - Medical devices (IV flow monitoring)

Conclusion:

Learned about various types of sensors, their purpose ,principle,technologies,applications and examples.